Concept:
This Board uses an GateMate FPGA
in combination with an USB PHY
and ab ISO 7816 Interface to
simulate an

ToDo Next Revision:
- Combine the two USB-C 5V LEDs into one
- BOM verkleinern, v.a. Kondensatoren
- Power IC mit mehreren Buck convertern
- PR2040 Flash did not work, replacement from Winbond did work

Done:
- Fixed 5V Diode
- Power supply structure reworken
- removed reset, header, 1nF caps
- replaced power sturcture
  -> buck for VDD_CORE
  -> adj Buck for IO voltages
  -> LDO for fixed 2V5 things: VDD_CLK, RAM
  -> LDO for fixed 1V8 things: SRAM
  -> LDO for fixed 3V3 things: RP2040USB
- Upgraded Interface to not reqire soldering to switch from master to slave and using a more common swich

FpgaMisc
File: FpgaMisc.kicad_sch

DirtyJtag
File: DirtyJtag.kicad_sch

FpgaIOs
File: FpgaIOs.kicad_sch

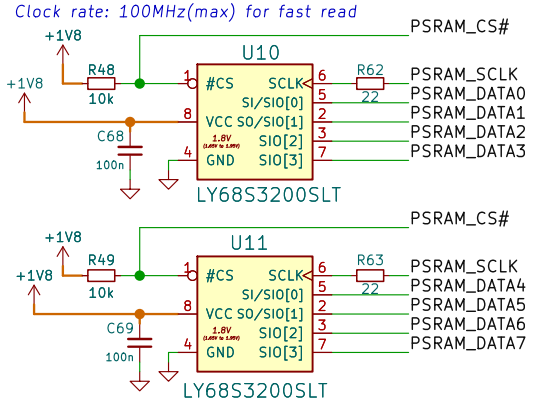
Power
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open source hardware

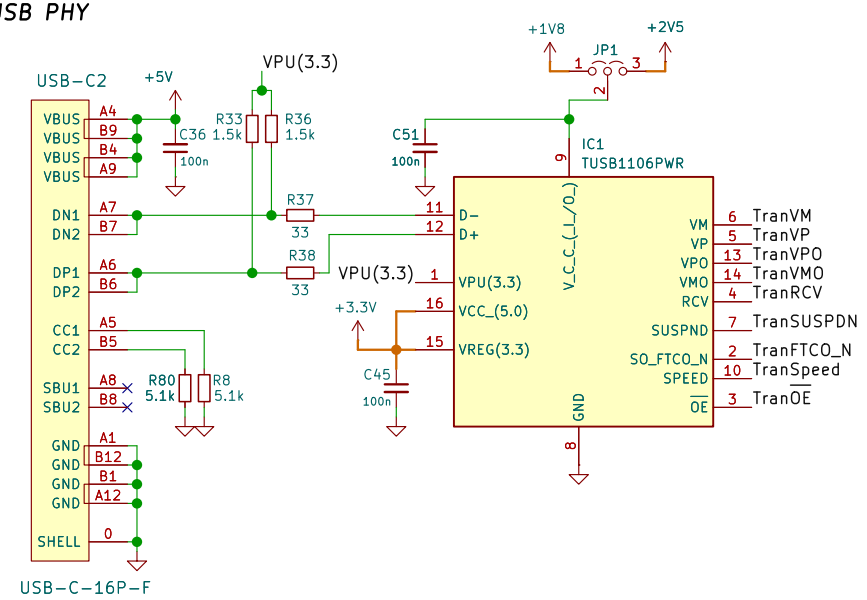
HM

Serial Pseudo-SRAM with QPI

Clock rate: 100MHz(max) for fast read



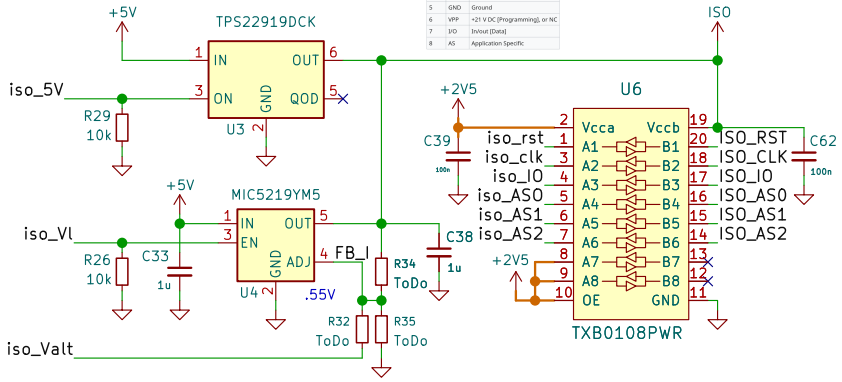
USB PHY



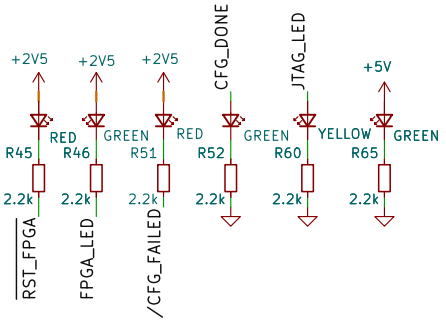
ISO/IEC 7816 Interface

Notes:
- Supports 3 classes: A,B,C
- Different Supply Voltages: 5V, 3.3V, 1.8V
- three AS - Application Specific pins
- does not support +21V DC [Programming]

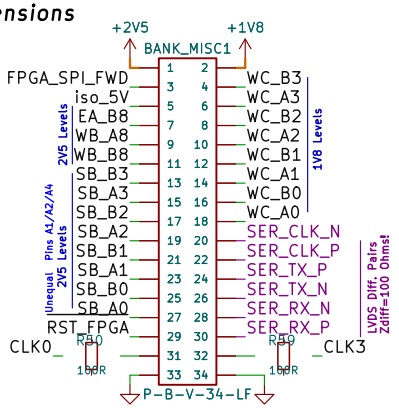
If used as Master:
- Place all components
- 5V direct pass though of the USB Voltage
- 3.3V, 1.8V 500mA current limited



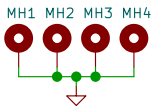
LEDs



Extensions



Mounting Holes



https://www.olimex.com/
based on GateMateA1-EVB

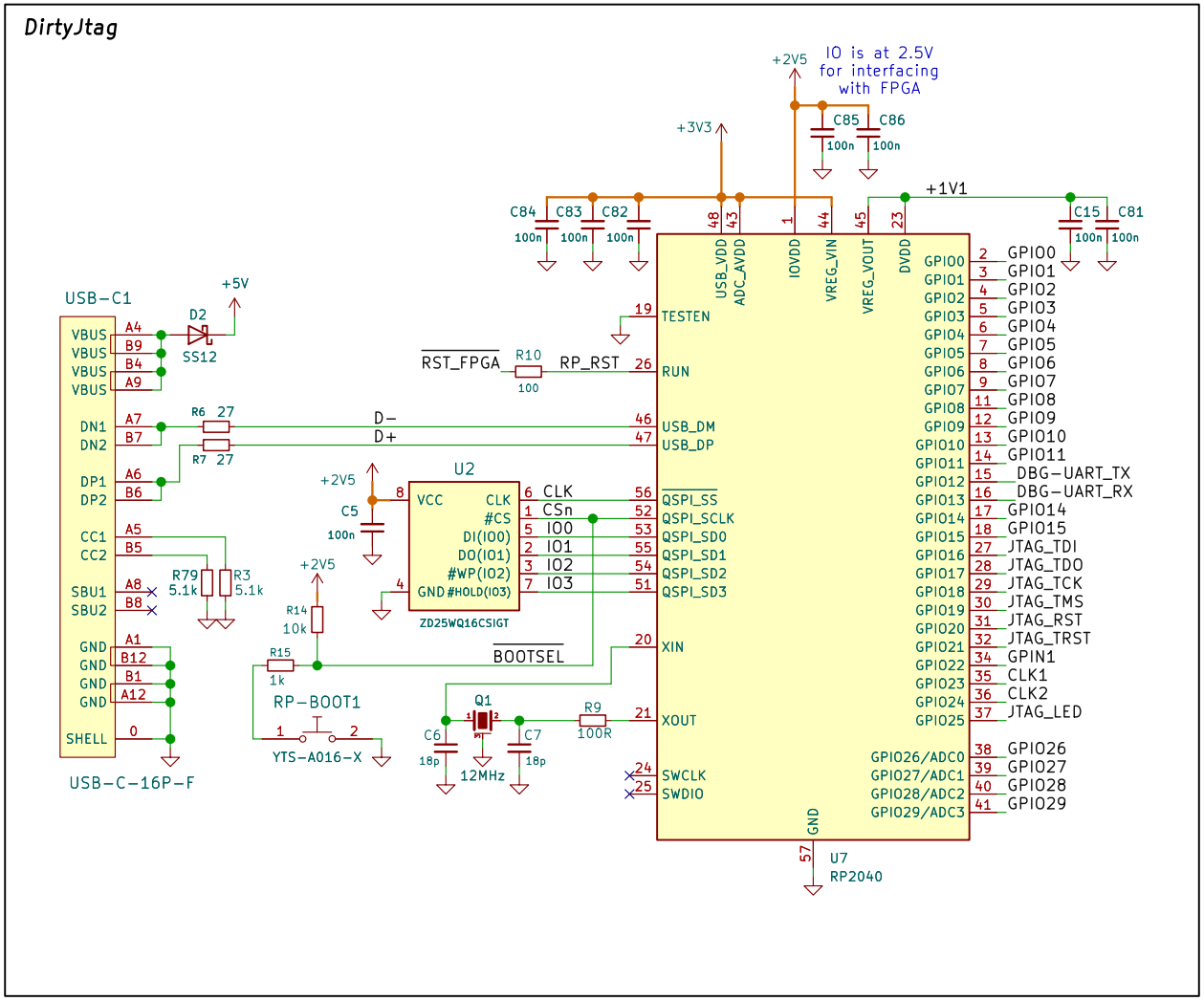
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Title: MillerV1

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Jtags

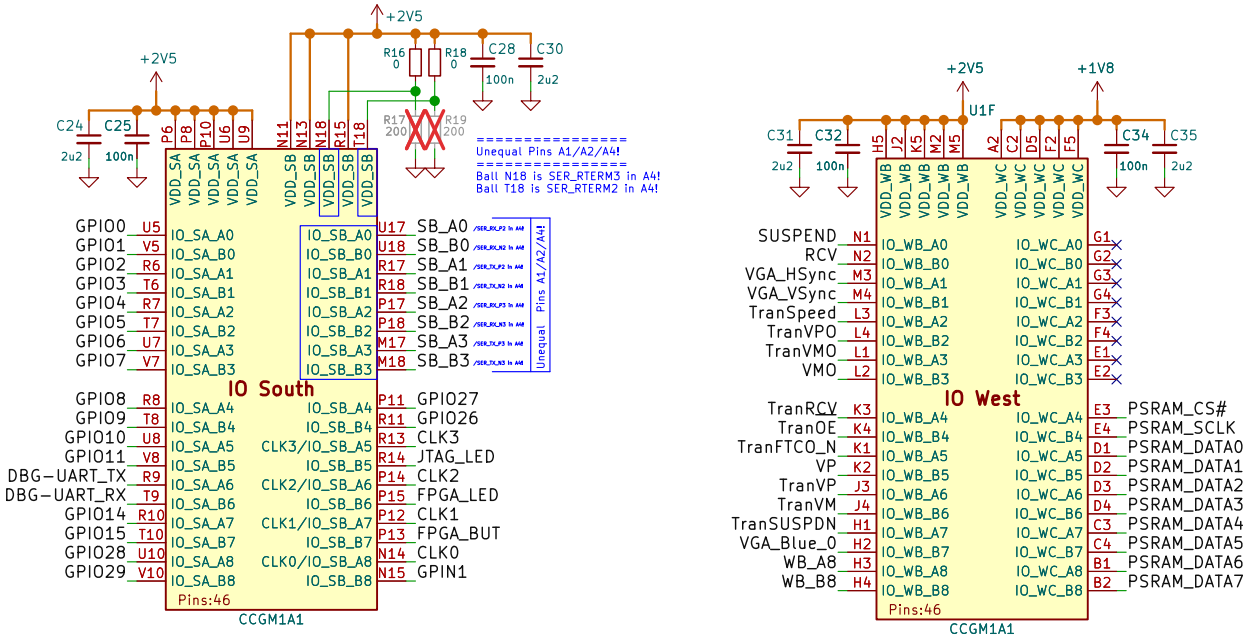
DirtyJtag

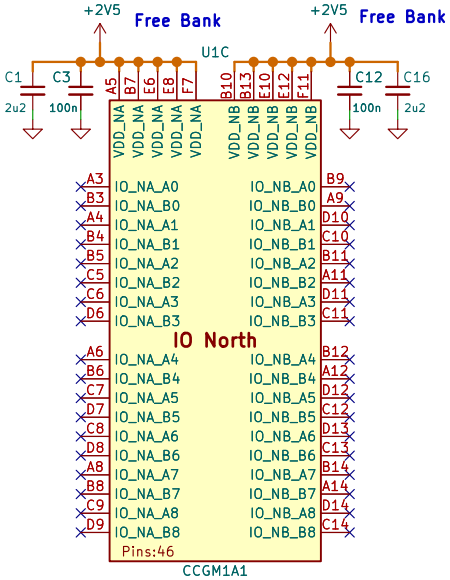


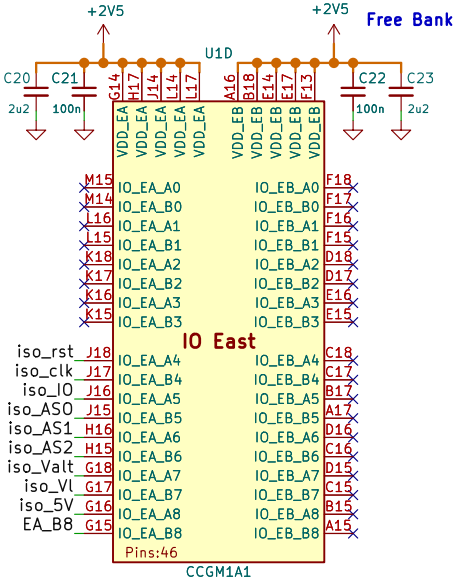
Notes:
1. RP2040 supports 5MHz to 15MHz crystals(XOSC) to produce an accurate reference clock(clk_ref). The reference clock is distributed to the PLLs, which can be used to multiply the XOSC frequency to provide accurate high speed clocks. For example, they can generate a 48MHz clock which meets the frequency accuracy requirement of the USB interface and a 133MHz maximum speed system clock. clk_ref = (6-12)MHz(Nominal) - Reference clock that is always running unless in DORMANT mode. Runs from Ring Oscillator (ROSC) at power-up, but can be switched to Crystal Oscillator (XOSC) for more accuracy.
2. There are two PLLs in RP2040. They are: pll_sys - Used to generate up to a 133MHz system clock(clk_sys), pll_usb - Used to generate a 48MHz USB reference clock(clk_usb).
3. System clock clk_sys = 125MHz(Nominal), Runs from clk_ref at power-up but is typically switched to a PLL.
4. Peripheral clock clk_peri = (12-125)MHz(Nominal). Typically runs from clk_sys but allows peripherals to run at a consistent speed if clk_sys is changed by software.
5. The maximum SPI clock frequency is half the system clock frequency(clk_sys). For example, at the maximum SSPCLK (clk_peri) frequency on RP2040 of 133MHz, the maximum peak bit rate in master mode is 62.5Mbps. In slave mode, the same maximum SSPCLK frequency of 133MHz can achieve a peak bit rate of 133 / 12 = ~11.083Mbps.
6. CLOCK GPINx - General purpose clock inputs. Can be routed to a number of internal clock domains on RP2040, e.g. to provide a 1Hz clock for the RTC, or can be connected to an internal frequency counter. GPIN0, GPIN1 and XIN inputs are limited to 50MHz.
7. CLOCK GPOUTx - General purpose clock outputs. Can drive a number of internal clocks (including PLL outputs) onto GPIOs, with optional integer divide. clk_gpout[0..3] - Clock output to GPIO. Can be used to clock external devices or debug on chip clocks with a logic analyser or oscilloscope.

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FPGA IOs







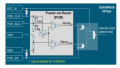
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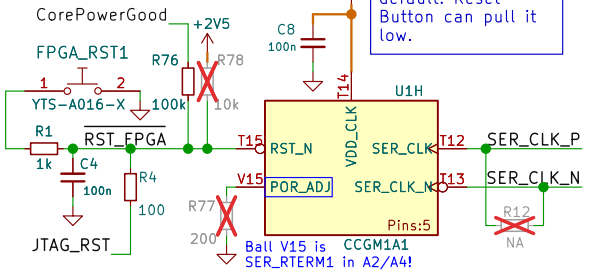
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KiCad E.D.A. 10.0.8		Id: 3/5

FPGA Misc

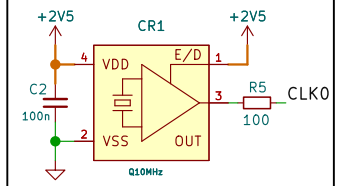
Reset / SerCLK



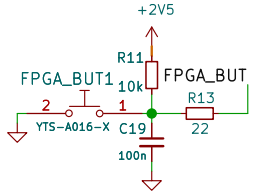
Powergood pulls RST_N high per default. Reset Button can pull it low.



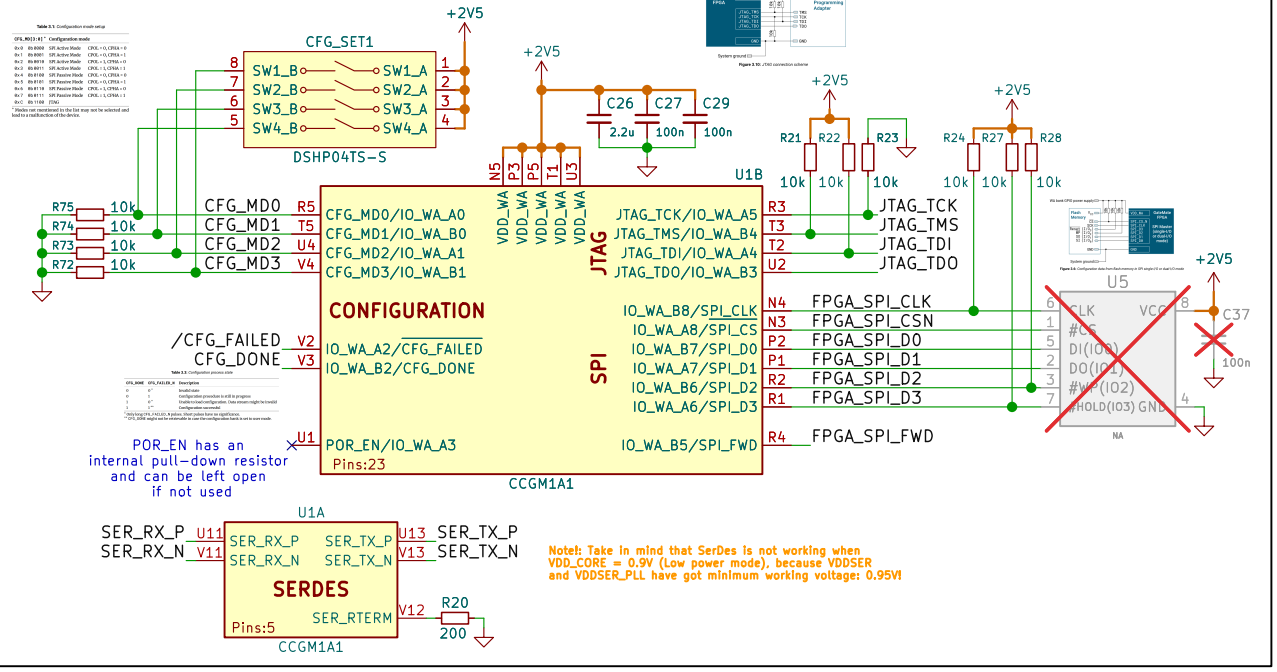
Clock



Button



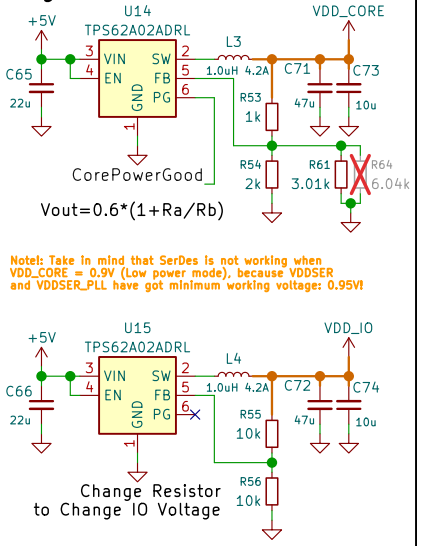
Flashing



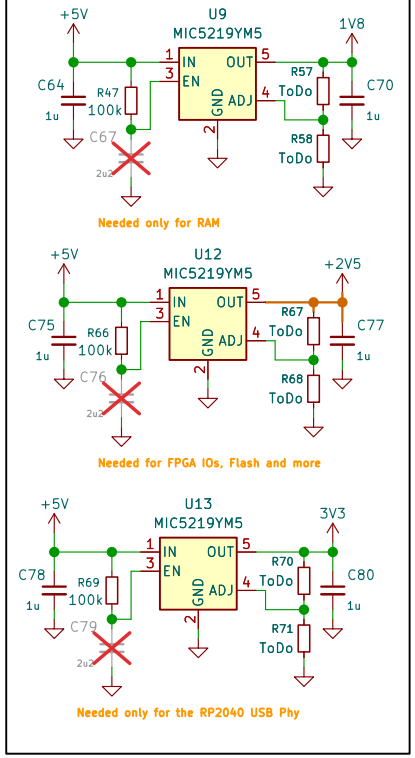
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Power

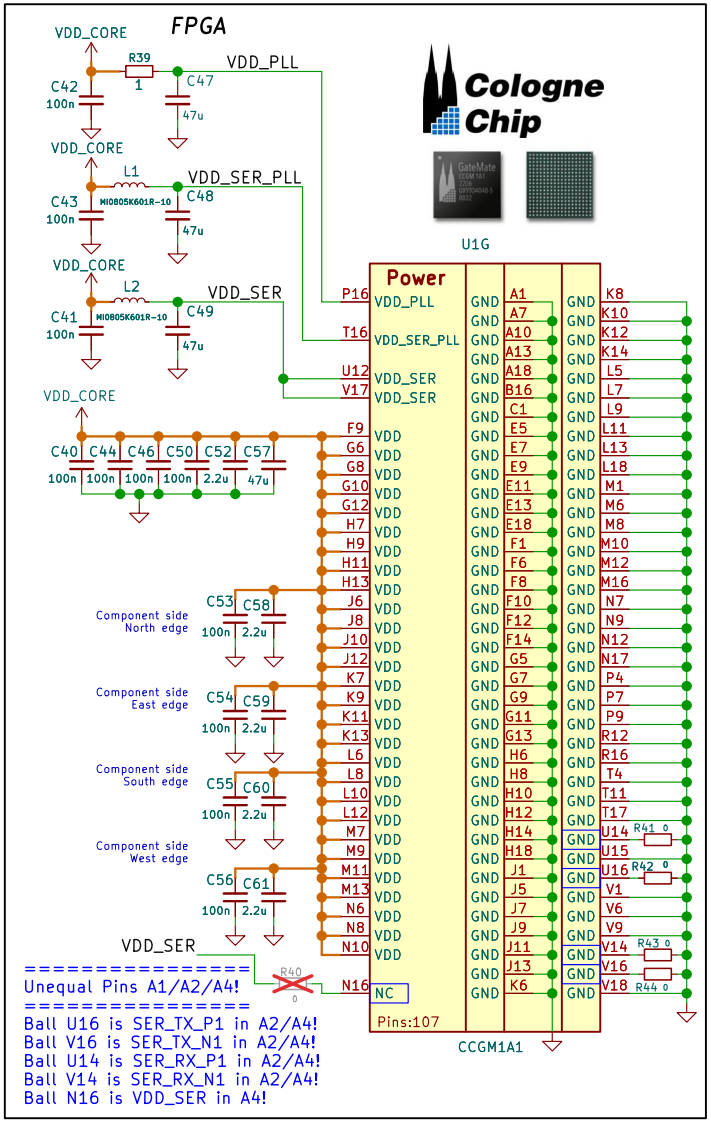
Voltage Converters FPGA



Voltage Converters Miscelanious



FPGA



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